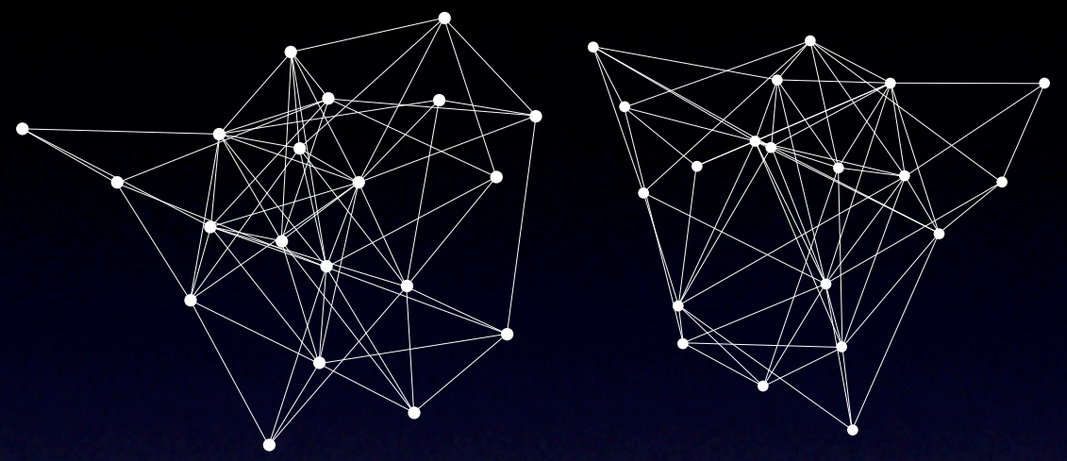
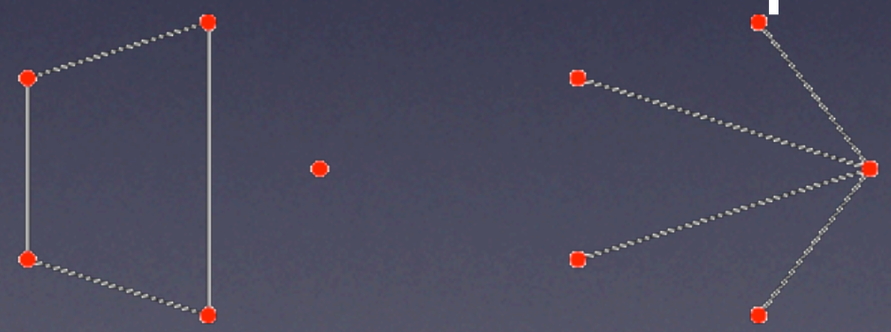
Graph Isomorphism

Are these  isomorphic graphs?

Good way to determine this in practice:
compare eigenvalues of adjacency matrices

But there are isospectral non-isomophic graphs:



eigenvalues: (-2,0,0,0,2) (-2,0,0,0,2)

Physicists Meet Graph Isomorphism

Adjacency matrix? Think random walk.

Two walkers? $A \rightarrow A \otimes I + I \otimes A$

no help

Two walkers that are bosons? $A \rightarrow P(A \otimes I + I \otimes A)P^\dagger$

$$P = \frac{1}{2}(I + SWAP)$$

no help

Two walkers that are fermions? $A \rightarrow P(A \otimes I + I \otimes A)P^\dagger$

$$P = \frac{1}{2}(I - SWAP)$$

no help

Physicists Meet Graph Isomorphism

Two walkers that are hard-core boson?
(bosons that can't sit on top of each other)

$$A \to P(A \otimes I + I \otimes A)P^\dagger - S$$

Physicist: seems to work for most examples I check!

CSer: doesn't work for strongly regular graphs

Physicist: try three walkers!

CSer: seems to work for strongly regular graphs we try

A bit of time passes

Physicists Meet Graph Isomorphism

CSer: on second thought, k hard-core boson walkers no more powerful than most classic CS graph isomorphism algorithm of th 70s+80s (the m-dimensional Weisfeiler-Lehman algorithm) and this algorithm is known to fail

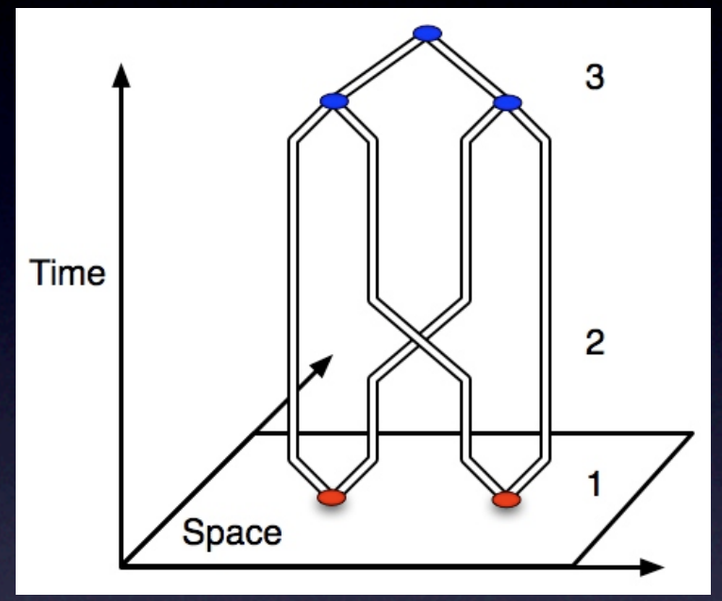
no help

bah hah hah



Dave Bacon Meets Graph Isomorphism

Can we use anyons?

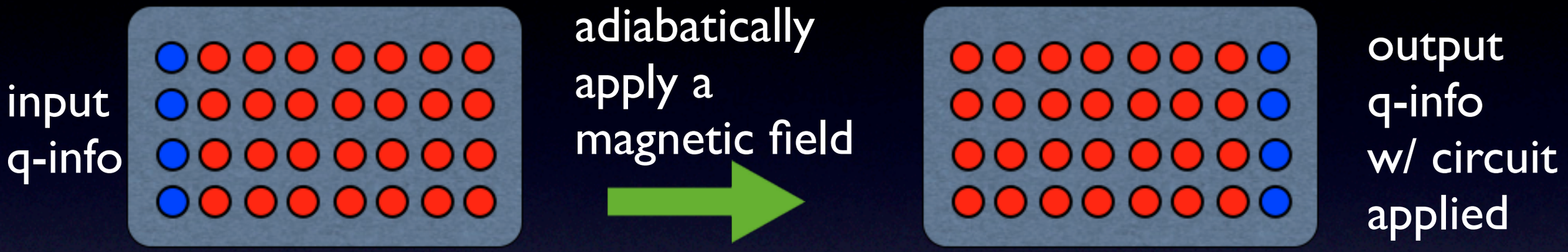


WTH are anyons?
- quasiparticles in fractional quantum hall effect (Nobel Prize for Laughlin)
- particles in topological quantum field theory that "explain" the Jones knot polynomial (Fields medal for Ed Witten)

If "results" followed "awards" we'd be done.
Instead we must work hard to understand how these topological invariants can or cannot help find efficient quantum or classical algorithms for GI

Other QuantumW Theory Group Work

Quantum Transistors



MUSIQC

~15 million dollar project to build at ~100 qubit ion trap quantum computer (Maryland, Duke, Michigan, UW, UBC, Georgia Tech)